

Software Testing

EvoSuite Testing Framework

João Saraiva

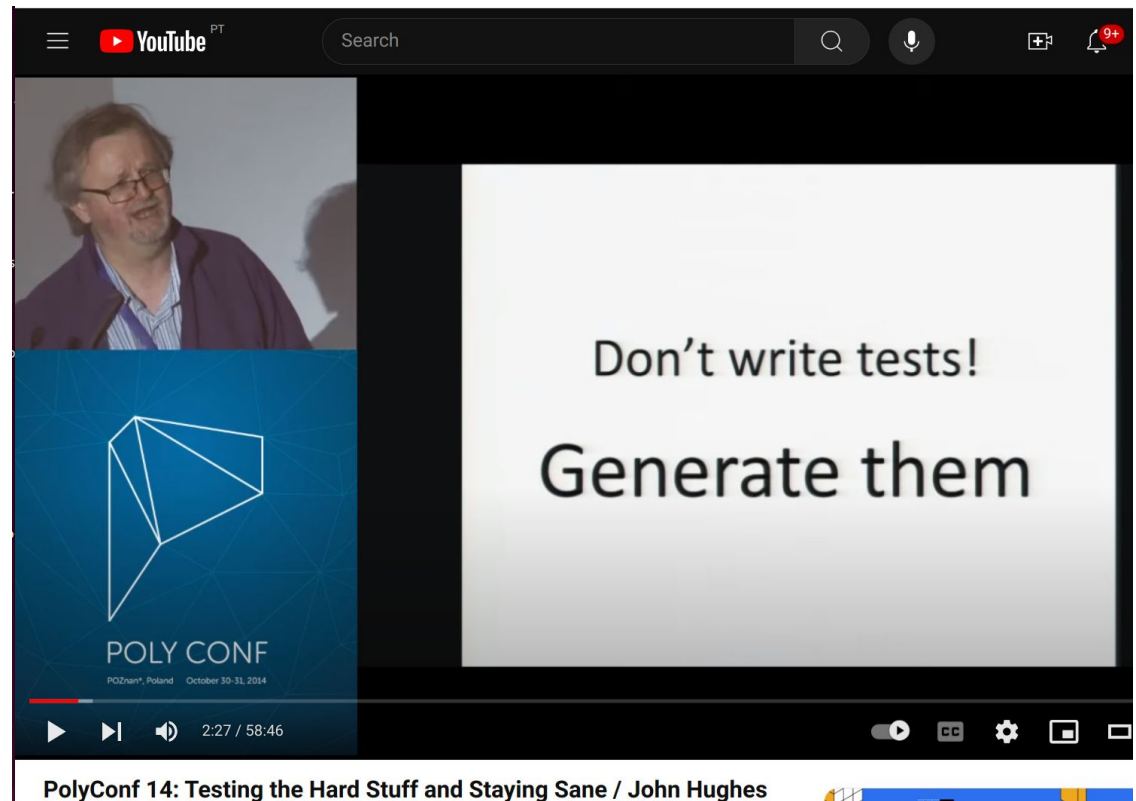
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***Departamento de Informática
Universidade do Minho, Portugal***

Don't write tests! Generate them

John Hughes:

- ▶ One of the principal designers of the **Haskell** programming language,
- ▶ the creator of **QuickCheck**, a foundational framework in the field of **property-based testing**.



[John Hughes - Don't Write Tests!](#)

Automated Unit Test Generation for Java

EvoSuite: Motivation

- Writing unit tests is time-consuming
- Manual tests often have limited coverage
- Automated testing can increase coverage and detect faults
- **Goal:** reduce developer effort in testing

EvoSuite: Automated Unit Test Generation

- Framework for automatic generation of Java unit tests
- **Produces JUnit test suites**
- Uses evolutionary algorithms
- Targets coverage goals such as branch coverage

Search-Based Software Testing

- Testing formulated as an optimization problem
- Goal: maximize code coverage
- Uses genetic algorithms
- Tests evolve over multiple generations

Fitness Functions

- Search guided by a fitness function
- Common objectives:

Branch coverage

Line coverage

Exception coverage

Mutation score

Generated Test Suites

- Output consists of JUnit test classes
- Includes automatically generated assertions
- Tests are minimized and simplified
- Readable by developers

Tool Integration

- Command-line interface
- Integration with Maven
- IDE support (Eclipse, IntelliJ)

- Example command:

```
java -jar evosuite.jar -class MyClass
```

Advantages

- High code coverage
- Fully automatic
- Detects real bugs
- Useful for testing legacy code

Limitations

- Complex object construction
- External dependencies
- Non-deterministic behavior
- Generation time for large systems